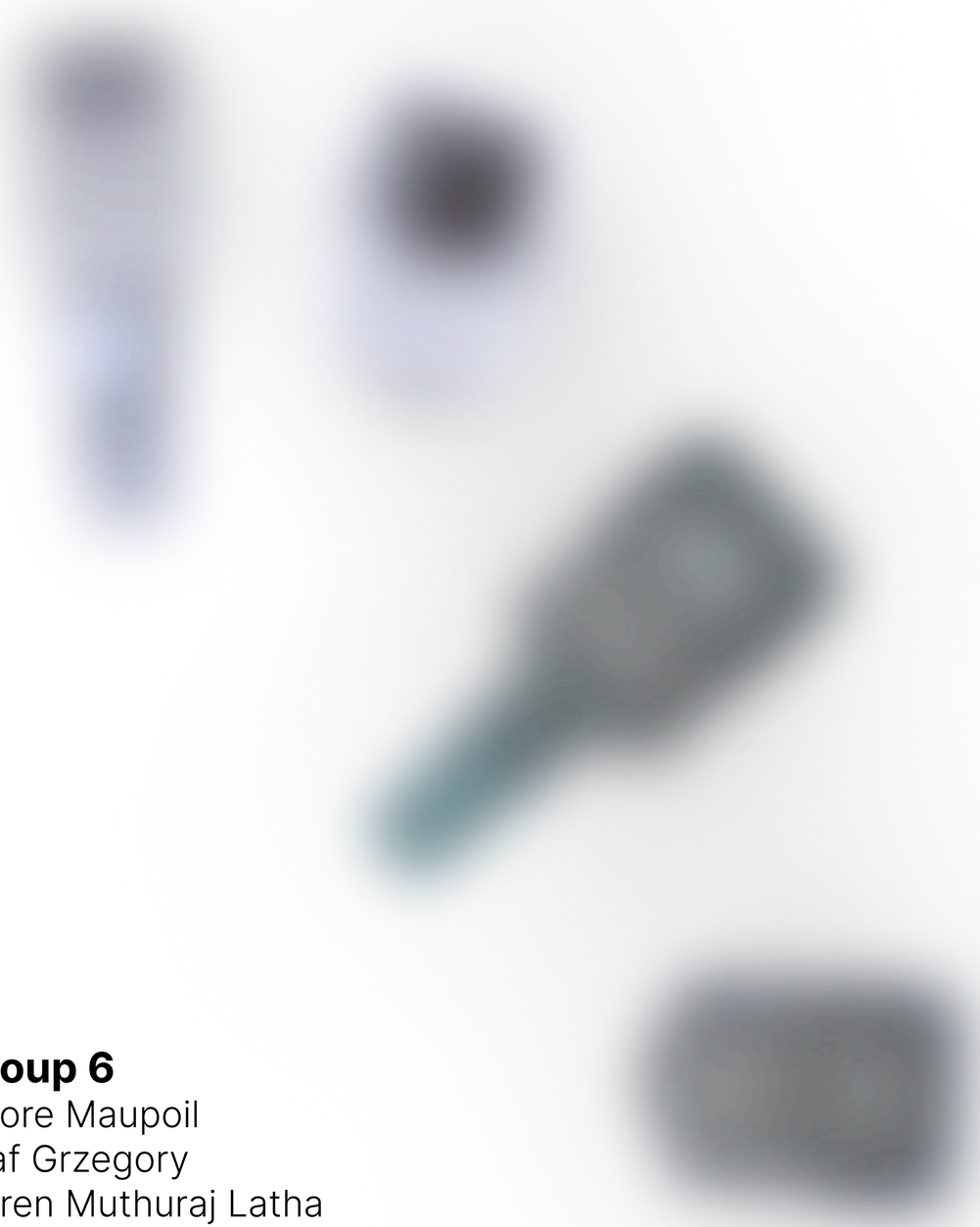


Product Development Studio 1
2025-26

Silvia Ferraris
Marco Meraviglia
Marina Carulli
Giorgio Colombo

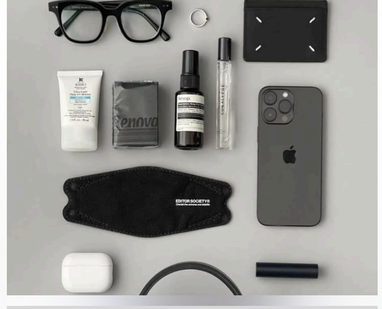
Nebu 90•

Nebu 90•

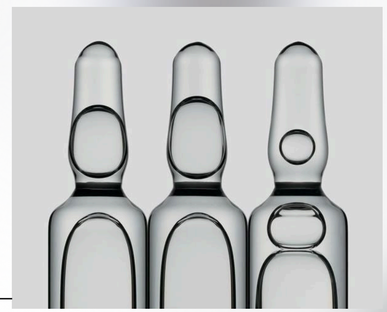
Group 6
Ettore Maupoil
Olaf Grzegory
Naren Muthuraj Latha
Martina Hrncirikova

POLITECNICO
MILANO 1863

CONTEXT
Research/User









Doctor & Pharmacist Interviews

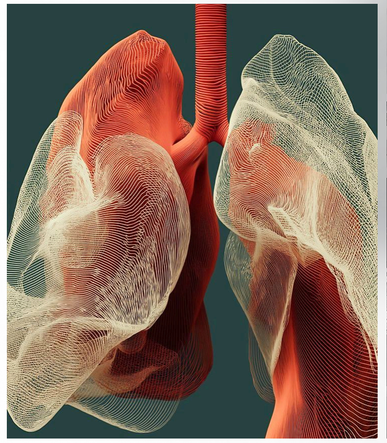
User

Chronic patients (especially asthma)

//Many chronic patients own two devices: a large home nebulizer and a **small portable** one for **daily activities**

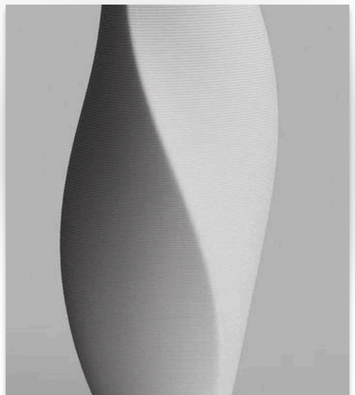
Needs

//Simple controls and **intuitive operation**
//**Compact size**
//Soft materials and comfortable **interfaces**
//Modern, **discreet** appearance
//Less medical feeling









Scenario

//People affected by **chronic respiratory conditions** (asthma, COPD or cystic fibrosis)

//Treatments at home, at work, school or while **traveling** without looking like a patient



COUNTER BRIEF & REQUIREMENTS

A compact, portable nebulizer designed for **discreet** everyday use by **chronic respiratory patients**. Efficient **5–10-minute** treatments anywhere through intuitive feedback, quiet sound, and subtle design empowering frequent use **without signaling medical dependency**.



Portability

Weigh around 200g and be compact enough to fit easily into a purse or pocket

Form and shape

Discreet and familiar design, more everyday personal object than a medical device

Battery

The battery should last at least 5 cycles of 15 min each for a full day of use

Ergonomics

Comfortable fit in hand, mask stays on the face without holding or adjusting

Hygiene

Mouthpiece should remain covered when the device is carried on the go

Interface

Higlighted buttons, feedback and interaction easily uderstandable



NATURAL ENVIRONMENTS

3 scenarios



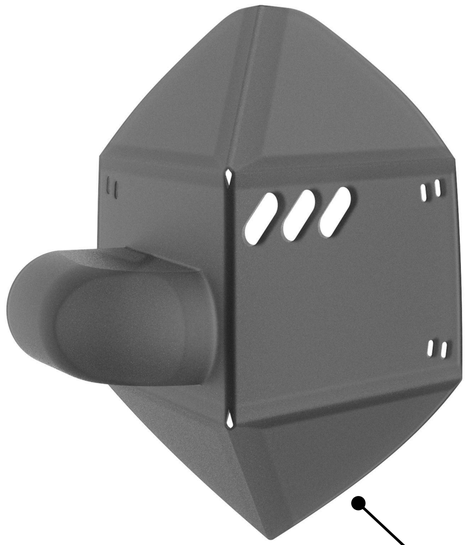




FEATURES
MAIN PARTS



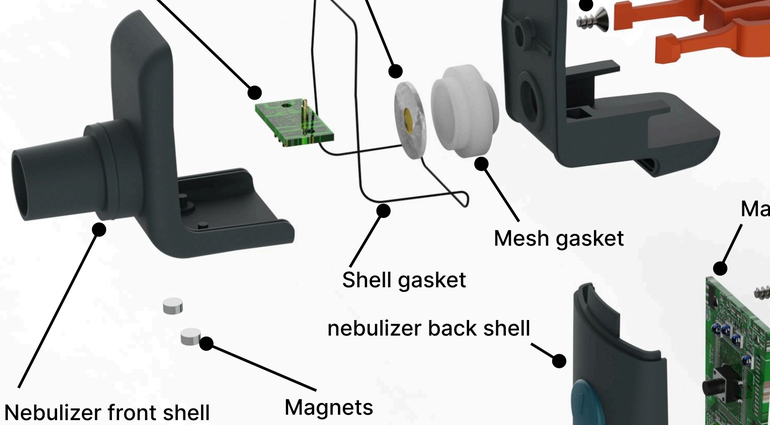
Cable to connect the main body with the nebulizer unite when in use with the mask



Foldable mask



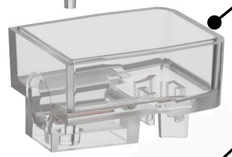
Mouth piece



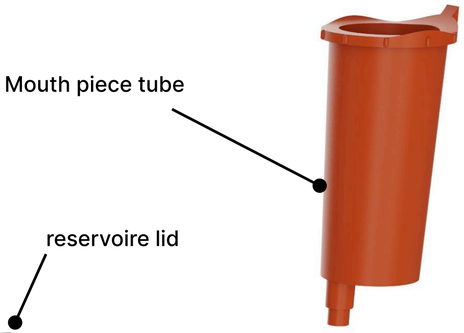
Pogo pin pcb
Vibrating mesh
Nebulizer back shell
M3×6mm screw
Nebulizer front shell
Magnets
Shell gasket
Mesh gasket



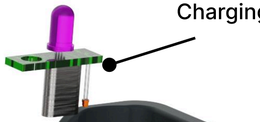
Water sensor pin



Reservoire bottom
Retainig clip



Mouth piece tube
reservoire lid



Charging Pcb



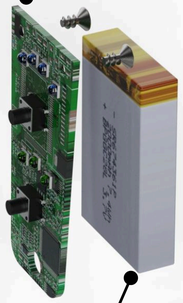
Top shell



Pogo pin pcb
Bottom shell



nebulizer back shell



Main pcb
1300 mAh battery



M3×16mm screw
Screw plug



M3×16mm screw

FEATURES
MECHANISMS

Pogo pins and magnet

Pogo pin and magnets allow in the system manteining a reliable connection



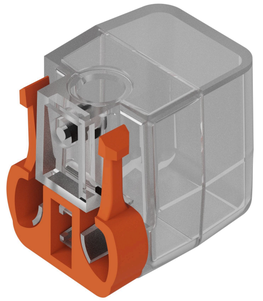


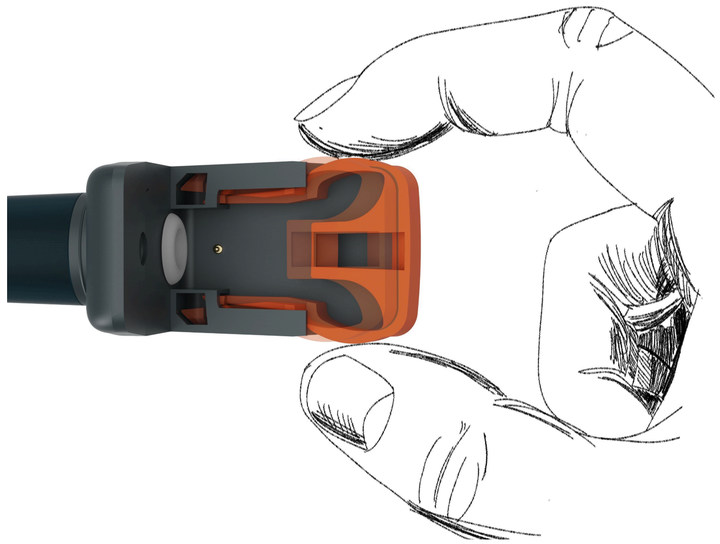
Foldable mask

the mask has a foldable and sleek design shifting weight from front to back and enhancing portability

Reservoire

The reservoire has 15ml of capacity and is secured to the body thanks to a modified buckle design





INTERACTIONS & FEEDBACK

Power

Pressing repeatedly to cycle through 3 settings

Low

Mid

High





Timer

Pressing repeatedly to cycle through 3 settings

5 min

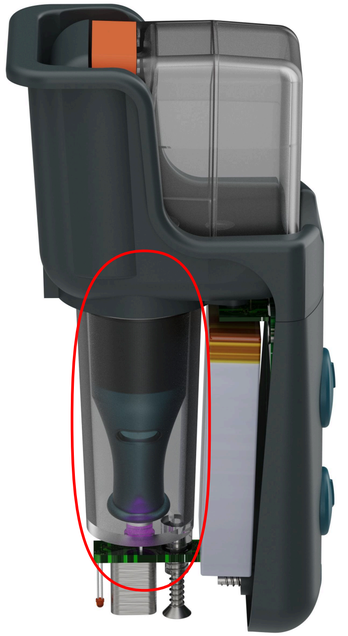
10 min

15 min

Self cleaning

The device is equipped with self cleaning function, for enhanced safety it will turn on only while charging

3.7V 254 nm



STORYBOARD

Preparation + mode of use **01**



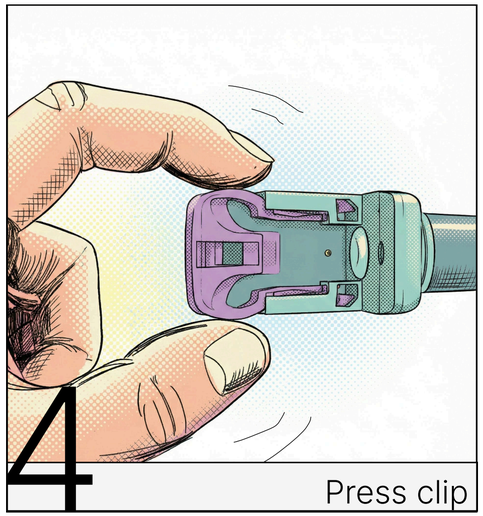
1 Access the device



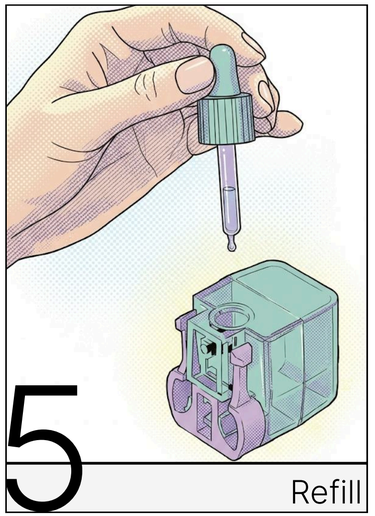
2 Prepare



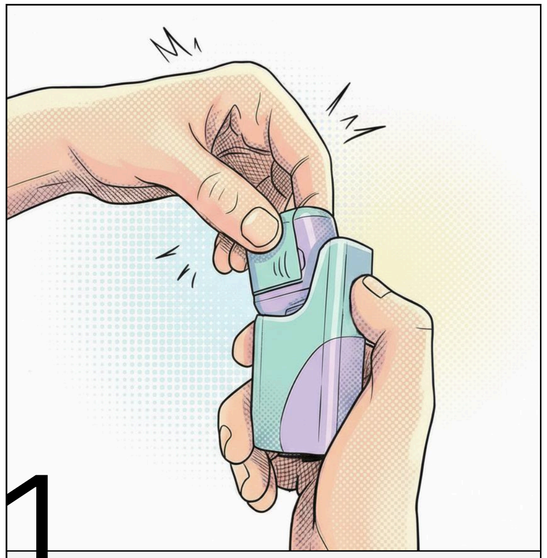
3 Release the medicine tank



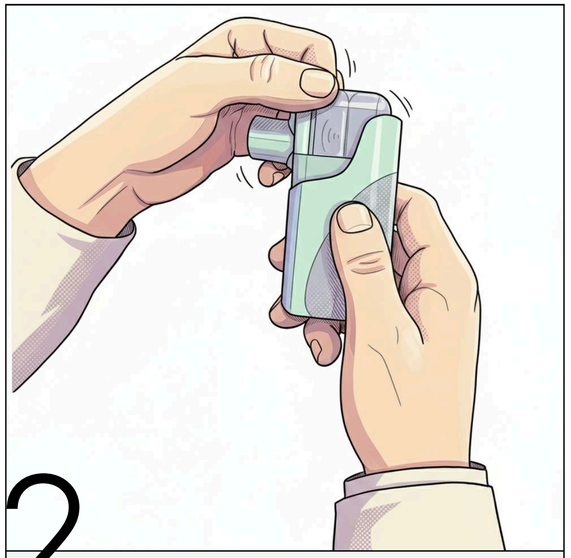
4 Press clip



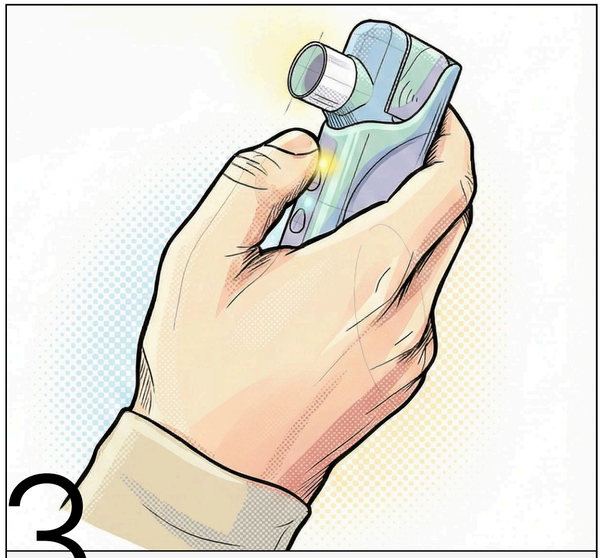
5 Refill



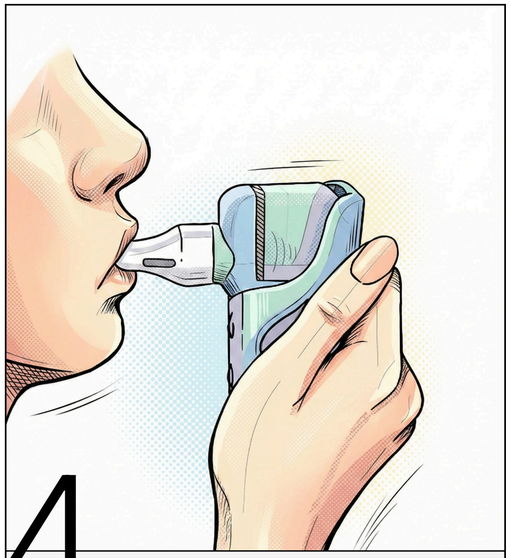
1 Detach the head unit



2 Flip it and place it back



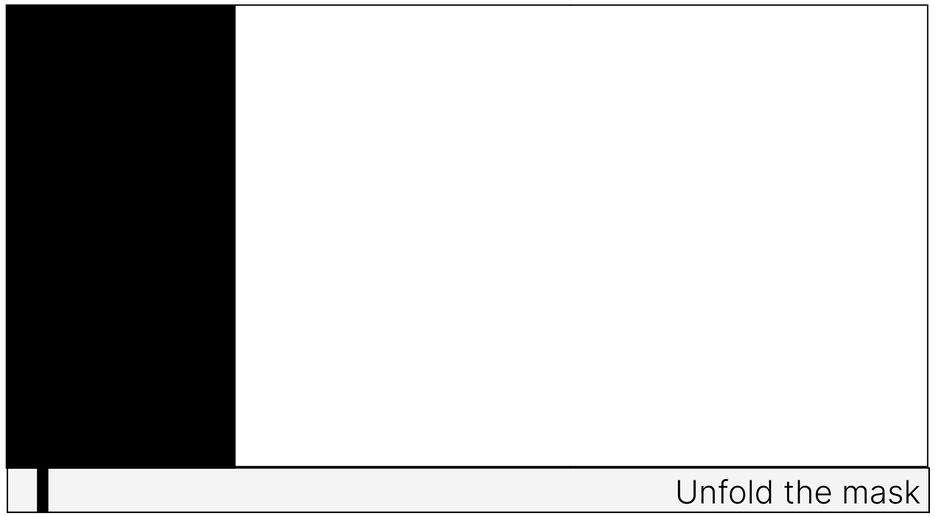
3 Select the mode



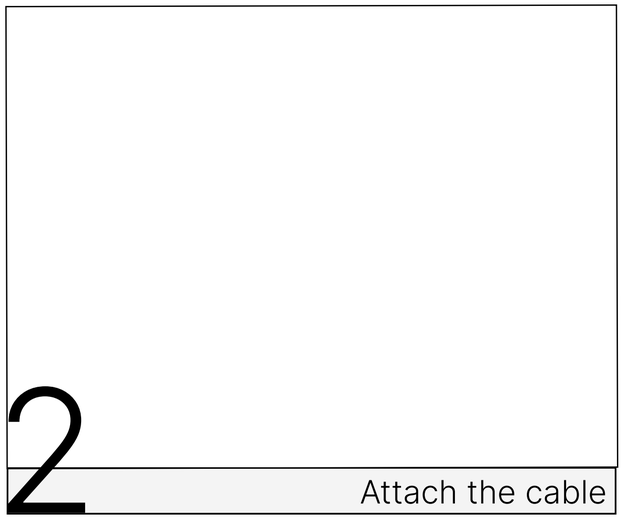
4 Inhale

STORYBOARD

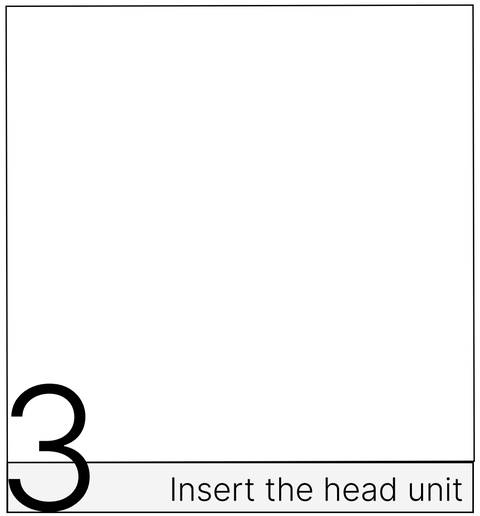
Mode of use **02**



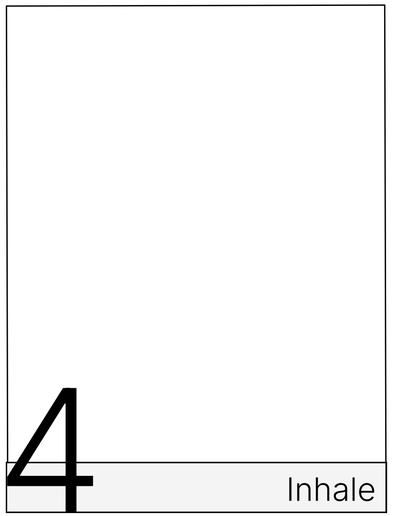
1 Unfold the mask



2 Attach the cable

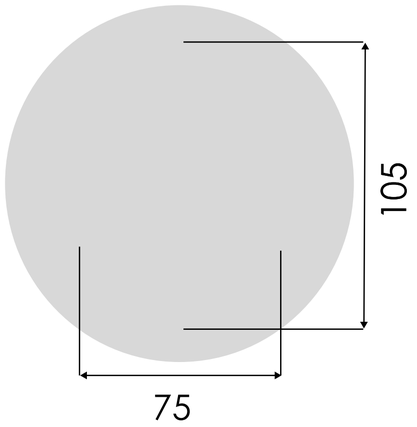


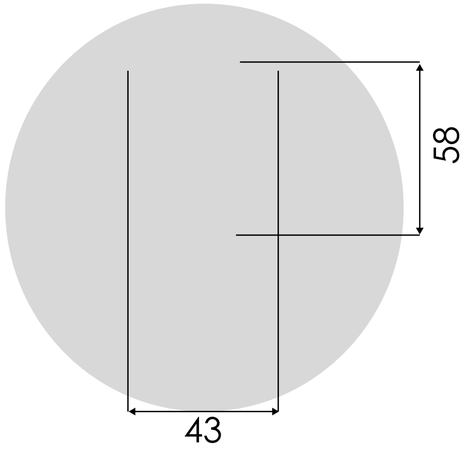
3 Insert the head unit

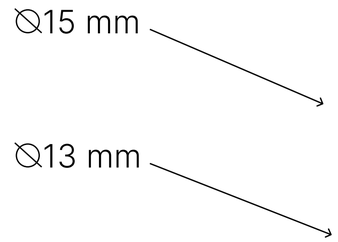


4 Inhale

ERGONOMICS







TECHNICAL FEATURES

NEBULIZER

Shell: ABS
Buttons: overmolded silicone

TECHNICAL FEATURES

MASK

Mask: ABS

THANK YOU
GRAZIE